Wonjun Lee

Assistant Professor at OSU

Columbus, OH

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Research Interests

My research focuses on developing partial differential equations (PDE)-based algorithms to solve high-dimensional machine learning problems and analyze the theoretical properties of the algorithms.

Keywords: Applied math, machine learning, deep learning, generative modeling, contrastive learning, optimal transport, gradient flows, mean field games.

Academic Positions

The Ohio State University — *Columbus, OH*
Assistant Professor — *Aug 2025 - Present*

- Assistant professor at the department of mathematics.

University of Minnesota, Twin Cities — *Minneaplis, MN*
IMA-NIST Postdoctoral Fellow — *Aug 2022 - Aug 2025*

- A joint IMA-NIST Postdoctoral Fellowship in Analysis of Machine Learning at the Institute for Mathematics and its Applications (IMA) in the College of Science and Engineering at the University of Minnesota (UMN).
- Working on machine learning research with Prof. Jeff Calder, Prof. Li Wang, and Prof. Gilad Lerman.

University of California, Los Angeles — *Los Angeles, CA*
Assistant Adjunct Professor — *Jun 2022 - Aug 2022*

- Taught introductory programming course in C++ (PIC 10A) as a main instructor.

Education

University of California, Los Angeles — *Los Angeles, CA*
Ph.D. in Mathematics — *Sep 2017 - Jun 2022*

- **Advisor:** Professor Stanley Osher.
- **Thesis:** Algorithms For Optimal Transport And Their Applications To PDEs.

George Mason University — *Fairfax, VA*
B.S. in Mathematics — *May 2015*

- Concentration in Applied Mathematics and Mathematical Statistics.
- GPA: 3.84/4.0 *magna cum laude*, Phi Beta Kappa.

Honors and Awards

2026 NVIDIA Academic Grant: GPU Computing Resources (Co-PI, Jan 2026 – Jun 2026)

2022 Rising Star in Data Science from the University of Chicago. PROFILE LINK.

2021 UCLA Dissertation Year Fellowship ($\$20,000$)

2014 2014 Outstanding Presentation Award at the Joint Mathematical Meetings, Baltimore, MD.

Service and Outreach

2024 Ambassador for the University of Minnesota's Data Science Initiative (DSI), **UMN DSI**

2024 Co-organizer of the minisymposium 'Theory and Applications of Optimal Transport in Machine Learning' with Jeff Calder, **SIAM Annual Meeting 2024**

Publications

- J. Calder, **W. Lee**, A.M. Neuman, *Spectral convergence rate of the graph p-Laplacian on a manifold with boundary*, **In preparation**
- M. Jacobs, **W. Lee**, *An efficient numerical scheme for tumor growth models*, **In preparation**
- J. Calder, **W. Lee**, *Dynamics Over Landscape: The Emergence of Linear Separability via Spectral Alignment in Contrastive Learning*, **Submitted**
- **W. Lee**, L. Wang, W. Li, *Generalized Deep JKO: an IMEX particle method for kinetic Fokker-Planck equations*, **Submitted**
- J. Gu, M. Mardani, **W. Lee**, D. Zou, G. Lerman, *Demystifying Latent Space Diffusibility via Fisher Geometry*, **Submitted**
- Y. Yang, J. Li, Y. Quan, D. Zou, **W. Lee**, X. Li, G. Lerman, *Anomaly Detection in Ancient Ceramics Using Gromov-Wasserstein Regularized Hyperbolic Autoencoders*, **Submitted**
- **W. Lee**, R. O'Neill, D. Zou, J. Calder, G. Lerman, *Geometry-Preserving Encoder/Decoder In Latent Generative Models*, **Submitted**
- **W. Lee**, Y. Yang, D. Zou, G. Lerman, *Monotone Generative Modeling via a Gromov-Monge Embedding*, SIAM Journal on Mathematics of Data Science, 2025
- J. Calder, **W. Lee**, *Monotone Discretizations of Levelset Convex Geometric PDEs*, Numerische Mathematik, 2024
- Y. Yang, **W. Lee**, D. Zou, G. Lerman, *Improving Hyperbolic Representations via Gromov-Wasserstein Regularization*, ECCV, 2024
- **W. Lee**, L. Wang, W. Li, *Deep JKO: Time-Implicit Particle Methods For General Nonlinear Gradient Flows*, Journal of Computational Physics, 2024
- **W. Lee**, S. Liu, W. Li, S. Osher, *Mean Field Control Problems For Vaccine Distribution*, Research in the Mathematical Sciences, 2022
- W. Li, **W. Lee**, S. Osher, *Computation Mean-Field Information Dynamics Associated With Reaction Diffusion Equations*, Journal of Computational Physics, 2022

- S. Agrawal, **W. Lee**, S. W. Fung, L. Nurbekyan, *Random Features for High-Dimensional Nonlocal Mean-Field Games*, Journal of Computational Physics, 2022
- A. Vepa, A. Choi, N. Nakhaei, **W. Lee**, N. Stier, A. Vu, G. Jenkins, X. Yang, M. Shergill, M. Desphy, K. Delao, M. Levy, C. Garduno, L. Nelson, W. Liu, F. Hung, F. Scalzo, *Weakly-Supervised Convolutional Neural Networks for Vessel Segmentation in Cerebral Angiography*, Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2022
- **W. Lee**, W. Li, B. Lin, A. Monod, *Tropical Optimal Transport and Wasserstein Distances in Phylogenetic Tree Space*, Information Geometry, 2021
- M. Jacobs, **W. Lee**, F. Léger, *The back-and-forth method for Wasserstein gradient flows*, ESAIM: COCV, 27:28, 2021.
- **W. Lee**, S. Liu, H. Tembine, W.C. Li, S. Osher., *Controlling propagation of epidemics via mean-field games*, SIAM Journal on Applied Math, 2020
- **W. Lee**, R.J. Lai, W. Li, S. Osher., *Generalized unnormalized optimal transport and its fast algorithms*, Journal of Computational Physics, 2020
- H. Gao, **W. Lee**, W. Li, Z. Han, S. Osher, and H. V. Poor, *Energy-efficient Velocity Control for Massive Rotary-Wing UAVs: A Mean Field Game Approach*, IEEE Globecom, 2020.
- Y. Kang, S. Liu, **W. Lee**, W. Li, H. Zhang, and Z. Han, *Joint Task Assignment and Trajectory Optimization for a Mobile Robot Swarm by Mean-Field Game*, IEEE Globecom, 2020.

Teaching Experience

University of Minnesota *Minneapolis, MN*
Instructor *Aug 2022 – Present*

Spring 2025	**Math 2243**: Linear Algebra and Differential Equations
Fall 2024	**Math 2243**: Linear Algebra and Differential Equations
Summer 2024	**Summer School**: Machine Learning Summer Camp for high school students
Spring 2024	**Math 2243**: Linear Algebra and Differential Equations
Summer 2023	**Summer School**: Machine Learning Summer Camp for high school students
Summer 2023	**Summer School**: Random Structures in Optimizations and Related Applications
Spring 2023	**Math 2243**: Linear Algebra and Differential Equations

University of Minnesota *Minneapolis, MN*
Mentor from Directed Reading Program (DRP) *Aug 2023 - May 2024*

- Mentoring undergraduate students for the quarter-long independent study project in math.
- Topics: survey of optimization methods and their applications in neural newtorks.

University of California, Los Angeles *Los Angeles, CA*
Teaching Assistant *Aug 2017 – Jun 2021*

- PIC 10ABC: Intro, intermediate, advanced C++ programming.
- PIC 16: Python with Applications - Python modules such as PyQt, SciPy, Pandas, and NLTK.
- Math 164: Fundamentals of optimization. Linear / nonlinear programming.
- Math 151B: Applied numerical methods with analysis of algorithms and computer implementations.

University of California, Los Angeles *Los Angeles, CA*
Mentor from Directed Reading Program (DRP) *Jan 2021 - Mar 2022*

- Mentoring undergraduate students for the quarter-long independent study project in math.
- Topics: unsupervised learning of image segmentation, generative adversarial networks, applications of mean field games in finance.

Work Experience

University of California, Los Angeles *Los Angeles, CA*
Research Assistant *Aug 2017 – Aug 2022*

- Developed efficient algorithms using PDEs and optimal transport to solve challenging problems, including Wasserstein gradient flows, Navier-Stokes equations, and epidemic models. (PyTorch, C++)

George Mason University *Fairfax, VA*
Research Assistant *May 2017 – May 2018*

- Developed deep learning methods using SVD and diffusion map for classification tasks. (Tensorflow)

Cheiron, Inc *Washington D.C.*
Actuary *Feb 2015 - Sep 2016*

- Evaluated the likelihood of undesirable events using actuarial pricing and projection models.
- Worked on actuarial valuation reports for public, single-employer, and multi-employer plans.

Presentations, Talks

2026 2026 AI Research Summit, **The Ohio State University**
2025 Level Set Collective II, **UCLA**
2025 SIAM Central States Section Annual Meeting 2025, **Arkansas, US**
2025 PDE Seminar, **North Carolina State University**
2025 Computational and Applied Mathematics Seminar, **New Jersey Institue of Technology**
2025 Mathematics Colloquium, **Williams College**
2025 Research Seminar, **The Center for Communications Research-La Jolla**
2025 Applied Mathematics Seminar, **Tufts University**
2025 Recruitment Seminar, **The Ohio State University**

2024 UMTC-UMD Postdoc Seminar Program, **University of Minnesota, Duluth**

2024 (Poster) SIAM Conference on Mathematics of Data Science (MDS24), **Atlanta Georgia**

2024 SIAM Conference on Mathematics of Data Science (MDS24), **Atlanta Georgia**

2024 SIAM Annual Meeting (SIAM AN24), **Spokane, Washington**

2024 SE Machine Learning and Data Science Seminar, **University of Vienna**

2024 Computational and Applied Math (CAM) Seminar, **Georgia Tech**

2024 MDL Collective Seminar, **Iowa State University**

2024 Applied Math Seminar, **UC Santa Barbara**

2023 Workshop on Models and Algorithms for Path Planning (MAPP), **UT Austin**

2023 Analysis and Probability Seminar, **Iowa State University**

2023 A Monthly Seminar at The Mokaplan Research, Mokameeting, **Inria Paris**

2023 Kernel Club, **Colorado School of Mines**

2023 ACMD Seminar, **National Institute of Standards and Technology (NIST)**

2023 Workshop on kinetic and optimal transport, **University of Minnesota**

2023 2023 Algorithms for Threat Detection PI Workshop (Poster session), **Washington D.C.**

2023 The Level Set Collective II, **UCLA**

2023 SIAM Conference on Computational Science and Engineering , **CSE23**

2022 IMA Data Science Seminar, **University of Minnesota**

2021 Optimal transport and Mean field games Seminar, **University of South Carolina**

2021 Current Literature in Applied Mathematics Seminar, **UCLA**

2020 Optimal transport and Mean field games Seminar, **University of South Carolina**

2020 The Level Set Collective, **UCLA**

2019 Optimal transport and Mean field games Seminar, **UCLA**

2019 Optimal transport and Mean field games Seminar, **UCLA**

Skills and Hobbies

Programming C/C++, Python, Matlab
Language English, Korean
Hobbies Movies and video games